Subject:database management system

Total marks: 30

Part A [5 x 2 = 10 Marks]

3.1. Is $A \rightarrow C$ direct or derived?

- In the relation R(A, B, C) with dependencies $\{A \rightarrow B, B \rightarrow C\}$, the dependency $A \rightarrow C$ is **derived through transitivity**.
- According to **Armstrong's Axioms (Transitive Rule)**, if $A \rightarrow B$ and $B \rightarrow C$ hold, then $A \rightarrow C$ must also hold.

3.2. Prime vs. Non-prime Attribute

- **Prime Attribute:** An attribute that is a member of any candidate key of the relation.
  - *Example:* In a table Student(ID, Name) where ID is the primary key, **ID** is a prime attribute.
- **Non-prime Attribute:** An attribute that is not part of any candidate key.
  - *Example:* In the same table, **Name** is a non-prime attribute.

3.3. Closure of the set {A}

- Given $F = \{A \rightarrow B, A \rightarrow C, AB \rightarrow D\}$:
  1. Start with $A^+ = \{A\}$.
  2. Using $A \rightarrow B$, $A^+ = \{A, B\}$.
  3. Using $A \rightarrow C$, $A^+ = \{A, B, C\}$.
  4. Using $AB \rightarrow D$ (since both A and B are in the set), $A^+ = \{A, B, C, D\}$.
- **Answer:** $\{A\}^+ = \{A, B, C, D\}$.

3.4. Lossless Join and Normalization

- A **Lossless Join** ensures that when a relation is decomposed into two or more smaller relations, joining them back together results in exactly the original relation without any "spurious" (extra/fake) tuples.
- **Significance:** It is a required property for a successful normalization process to ensure data integrity.
- *Example:* If $R(A, B, C)$ is decomposed into $R1(A, B)$ and $R2(A, C)$, the join is lossless if A is a key for $R1$ or $R2$.

3.5. Join Dependency and 5NF

- **Join Dependency:** A relation R satisfies a join dependency if it can be recreated by joining several of its projections, even if no multivalued dependencies exist.
- **Relation to 5NF:** A table is in **5NF (Project-Join Normal Form)** if it is in 4NF and every join dependency in the table is implied by the candidate keys.
- *Example:* A table linking (Supplier, Part, Project) where a supplier provides a part only for a specific project might need to be split into three tables to avoid redundancy.

Part B [2 x 10 = 20 Marks] (either a or b)

3.6. a. University Database Normalization (to 3NF)

- **Attributes:** {StudentID, CourseID, InstructorName, Department, CourseName}.
- **Functional Dependencies:**
    1. $\{StudentID, CourseID\} \rightarrow \{InstructorName, Department, CourseName\}$ (Full Key Dependency).
    2. $\{CourseID\} \rightarrow \{InstructorName, Department\}$ (Partial Dependency).
- **Step 1 (1NF to 2NF):** Remove **Partial Dependency**.
    - Table 1: Enrollment(StudentID, CourseID, CourseName).
    - Table 2: Course_Details(CourseID, InstructorName, Department).
- **Step 2 (2NF to 3NF):** Check for **Transitive Dependencies**. If InstructorName determines Department, split further.
    - Final Schema: Enrollment(SID, CID), Course(CID, CName), Staff(Instructor, Dept).

3.6. b. [OR] Orders Table Normalization (to 3NF)

- **Current State:** {OrderID, ProductID, CustomerName, ProductName, Quantity}.
- **Dependencies:**
    1. $OrderID \rightarrow ProductID$.
    2. $ProductID \rightarrow ProductName$.
    3. $\{OrderID, ProductID\} \rightarrow Quantity$.
- **Normalization Steps:**
    - **2NF:** The dependency $ProductID \rightarrow ProductName$ is a partial dependency if $\{OrderID, ProductID\}$ is the composite key.

    - **3NF:** $OrderID \rightarrow ProductID \rightarrow ProductName$ is a **Transitive Dependency**.
- **Final Tables:**
    - Orders(OrderID, CustomerName).
    - OrderItems(OrderID, ProductID, Quantity).
    - Products(ProductID, ProductName).

3.7. a. Converting 3NF to 4NF

- **The Problem:** 4NF addresses **Multivalued Dependencies (MVD)**, where one attribute determines a set of values in another attribute, independent of other attributes.
- **Hypothetical Example:** A Teacher table with Subject and Hobby. One teacher teaches multiple subjects and has multiple hobbies.
- **Before (3NF):** One table Teacher(TID, Subject, Hobby) causes massive redundancy because every subject must be paired with every hobby for that teacher.
- **After (4NF):** Decompose into two tables: Teacher_Subjects(TID, Subject) and Teacher_Hobbies(TID, Hobby).

3.7. b. [OR] Significance of 5NF

- **Significance:** 5NF handles cases where information can be reconstructed from smaller pieces but cannot be represented by simpler binary (two-column) relationships.
- **Transformation:** To move from 4NF to 5NF, you identify **Join Dependencies**.
- **Example:** A Sales table linking Agent, Company, and Product. If an agent represents a company AND sells a specific product AND that company makes that product, the data must be split into three tables: (Agent, Company), (Company, Product), and (Agent, Product) to eliminate redundancy.